SMART Design for Multi-Turn Medical Agents

Concrete Research Plan

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1. Problem Formulation

现状问题：当前所有medical chatbot evaluation（包括Nature Medicine的PreA RCT、PROSCA RCT、MedAgentBench等）都把agent当作单一不可分割的**intervention**——在enrollment时randomize一次（use chatbot vs. not），end-of-encounter测outcome。但真实的medical conversation是**sequential decision process**：

- Turn 1: 病人说"I have chest pain" → agent要选：ask about radiation? ask about duration? order ECG? escalate?
- Turn 2: 病人回答之后 → 下一个决策点
- ...
- Turn T: 给diagnosis / referral / 处方建议

每个decision point都是一个policy choice，整个encounter的outcome取决于这一串sequential decisions的组合。**Single-stage randomization**估计的是**"**整个**policy vs.** 什么都不用**"**的**ATE**，但告诉不了我们哪个子决策重要——这正是SMART（Sequential Multiple Assignment Randomized Trial, Murphy 2005）解决的问题。

Core research question：How do we identify the optimal **dynamic treatment regime (DTR)** for a medical chatbot, i.e., a policy π(a_t | history_t) that maximizes clinical outcome, using a statistically principled randomized trial design?

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2. Key Contributions

1. **First SMART-style evaluation framework for LLM medical agents**——把行为医学里成熟的DTR方法论搬到LLM eval
2. **Identifiability theory**——形式化什么条件下从multi-stage randomization可以identify optimal DTR（关键：sequential ignorability在LLM context下的含义 ）
3. **Estimation algorithms**——adapt Q-learning和A-learning到LLM action spaces（离散决策点上的stage-wise policy learning）

4. **Benchmark + empirical evidence**——一个可复现的simulation env + 真实/semi-synthetic data上的validation
5. **Practical playbook**——对社区开放的SMART protocol template，让其他组能跑类似trial

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3. SMART Design Specification

3.1 定义Decision Points（最关键的设计选择）

不是每个turn都是decision point。Decision points应该是**policy-relevant, discrete, prospectively-identifiable**的时刻。建议3个stage：

Stage 1: Information gathering strategy（对话第1-3轮）

- Arm A1a: Comprehensive screening（问完整review of systems）
- Arm A1b: Focused chief-complaint-driven（只深挖主诉）
- Arm A1c: Risk-stratified（先问red flags）

Stage 2: Tool/RAG invocation（在得到足够信息后）

- Arm A2a: Always retrieve guidelines
- Arm A2b: Conditional retrieval（基于uncertainty）
- Arm A2c: No retrieval（pure parametric knowledge）

Stage 3: Output strategy（给病人结论时）

- Arm A3a: Single most-likely diagnosis + plan
- Arm A3b: Differential + shared decision-making
- Arm A3c: Escalate to human clinician

→ Total: 3×3×3 = 27 possible DTRs (但SMART只需randomize到stage不需要所有组合)

3.2 Re-randomization规则

SMART的核心是**response-based re-randomization**。定义一个intermediate outcome R_t（对话中可测的"是不是走在对的路上"）：

- R_1 = 信息充分性分数（是否收集到了diagnosis所需的关键info，用rubric-based LLM judge自动scoring）
- R_2 = 诊断confidence calibration（agent报告的confidence vs. 实际答对的概率）

Randomization logic:

- All subjects randomized to Stage 1 arm uniformly

- After Stage 1: responders ($R_1 \geq$ threshold) → randomize to {A2a, A2b}; non-responders → randomize to {A2a, A2c} (skip RAG-conditional since info不够)
- Stage 3类似

这种response-adaptive re-randomization正是SMART区别于factorial的地方——让你学到"如果**Stage 1**成功，**Stage 2**应该怎么做；如果**Stage 1**失败，**Stage 2**应该怎么做"。

3.3 Primary Endpoint

Primary: Diagnostic accuracy at end of encounter (binary, vs. ground-truth diagnosis)
Secondary:

- Safety events (hallucinated dangerous advice, missed red flags — 用pre-specified rubric)
- Conversation efficiency (# turns)
- Simulated patient-reported satisfaction
- Calibration error

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4. Implementation: 这怎么真做出来

4.1 Patient Simulator（不用真病人，避免**IRB**和**scale**问题）

用已有资源：

- **MedQA / NEJM case reports** 作为ground truth case bank（~5000 cases with confirmed diagnosis）
- **Patient agent**：用Claude或GPT-4做standardized patient，prompt按照MedDialogRubrics / PATIENT-Ψ风格——只给它atomic medical facts，加"anti-hallucination guidance"防止穿帮
- **Validation of simulator**：跑一个small human study（10-20个临床医生），看他们能不能区分模拟对话vs真对话，agreement on diagnosis应该和真实case相近

为什么可行：MATRIX paper已经跑了2100 simulated dialogues；PATIENT-Ψ被用在MH training；技术路径是现成的。

4.2 Agent Under Test

至少3-5个configurations（SMART的目的就是比较policy，不是比较base model）：

- GPT-4o, Claude Sonnet 4.5, Llama-3.3-70B, Med-PaLM-class open model
- 每个base model × 上述Stage 1/2/3的arms = 实际需要implement的scaffolds

Tech stack:

- LangGraph or a simple state machine控制decision points
- Tools: PubMed search, UpToDate-style retrieval, calculator, ICD lookup
- 每个arm就是一个不同的prompt template + tool access pattern

4.3 Trial Execution

Sample size calculation (这是方法论paper必须做扎实的):

- 基于Murphy (2005) SMART sample size formula
- Detect 5% absolute difference in diagnostic accuracy between best vs. worst embedded DTR at $\alpha=0.05$, power=0.8
- 初步估算：~800-1500 simulated encounters per arm × 27 DTRs → total ~3000-5000 encounters (因为SMART的设计让每个subject贡献给多个DTR的estimate)

Randomization:

- Pre-register on OSF（虽然不是真人trial但posture要足）
- Use blocked randomization by case category (cardiology, GI, etc.)
- Save full logs with pre-committed seeds for reproducibility

4.4 Analysis

主要**estimand**: optimal DTR and its value $V(\pi^*)$

Methods (at least两个互相check):

1. **Q-learning (backward induction)**:
    - 在Stage 3: 估 Q_3(H_3, A_3) 用supervised regression
    - 在Stage 2: V_3(H_3) = max_a Q_3(H_3, a); regress Q_2(H_2, A_2) on [H_2, A_2] with pseudo-outcome V_3
    - 在Stage 1: 同理
    - 每一stage的regression用gradient boosting或linear model（保持interpretability）
2. **A-learning (advantage-based, doubly-robust)**:
    - 估 contrast function C_t(H_t, A_t, A_t') = Q_t(H_t, A_t) - Q_t(H_t, A_t')
    - 有双重稳健性：propensity model或outcome model对一个就行
3. **Inference**: bootstrap for CI on $V(\pi^*)$; Hudgens & Laber方法处理non-regular inference（optimal policy value is non-smooth in data）

4.5 Validation / Sanity Checks

- **Oracle comparison**：在simulator里你知道真optimal policy，验证SMART恢复得准

- **Replicate PreA-style result**：你的方法能不能复现"chatbot使用 → reduced consultation time"这种大效应
- **Robustness**：perturb patient simulator temperature，看DTR稳不稳

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5. Theoretical Component

这篇paper不能纯empirical。至少一个定理：

Theorem (informal): Under sequential ignorability (trial randomization makes it hold by design), positivity (all arms get nonzero prob), and consistency, the Q-learning estimator $\hat{\pi}^*$ converges to π^* (true optimal DTR) at rate $\sqrt{(n/K)}$ where K is # of decision points, when outcome regression is correctly specified.

The LLM-specific twist: What happens when the "state" H_t (conversation history) is高维且被base LLM内部压缩？讨论**representation learning**假设下的identifiability——需要H_t的sufficient statistic被可观测的summary (rubric scores, embedding) capture。这里可以写一小段representation-based IV / sufficient summary的理论。

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6. Timeline

月份	任务
M1	Simulator build + validation (human study with 10 clinicians)
M2	实现3 stages × 3 arms × 4 base models的agent scaffolds
M3	Pilot trial (N=200), debug randomization and logging
M4	Full trial run (N≈4000 encounters) — 大概 $3-8k API budget
M5	Q-learning + A-learning analysis, write theory section
M6	Ablations, robustness, write-up
M7	Internal review, prereg update, submit

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7. Risks and Mitigations

Risk	Mitigation

Simulator不reliable → estimates不trustworthy	Human validation study + sensitivity analysis across simulator configs + semi-synthetic real-data extension (MIMIC或EHRSHOT dialogue)
27 DTRs太多，power不够	Pre-specify 6 "embedded DTRs" of primary interest；其他exploratory
Reviewer说"这只是fancy ablation"	Emphasize: re-randomization conditional on intermediate response是factorial做不到的；DTR identification有独特causal structure
"Why not just RL?"	核心区别：SMART给**finite-sample, trial-based, unbiased** estimate with CI；RL给policy but no statistical guarantee on value
Clinical co-author需求	现在就开始找一个emergency medicine或internal medicine attending合作，SMART analysis他们懂
